

Lecture 05 - Clipping

This lecture focuses on algorithms that remove or trim geometry against boundaries. Clipping is a geometric operation before rasterization, not a visibility test between overlapping objects.

Source chunk	course_text_parts/03_lectures/05_clipping.txt
Extracted slide pages	56
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Slide walkthrough

Each row gives a short reading comment for one extracted slide page. Use it to turn sparse slide bullets into a human-readable path through the lecture.

Page	Slide title	How to read it
1	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
2	Clipping Projected Geometry 1/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
3	Clipping Projected Geometry 2/2	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
4	5.1 Cohen-Sutherland Line Clipping	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
5	5.1 Cohen-Sutherland Line Clipping	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
6	Brute-Force Line Clipping 1/2	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
7	Brute-Force Line Clipping 2/2	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
8	Endpoint Analysis 1/2	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.

Page	Slide title	How to read it
9	Endpoint Analysis 2/2	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
10	Binary Coding of Half-Planes 1/3	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
11	Binary Coding of Half-Planes 2/3	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
12	Binary Coding of Half-Planes 3/3	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
13	Pseudo Code	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
14	Example	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
15	Evaluation	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
16	5.2 Cyrus-Beck Line Clipping	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
17	Cyrus-Beck Line Clipping	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
18	Parametric Line Representation	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
19	Intersection Point Calculation 1/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
20	Intersection Point Calculation 2/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
21	Determine Relevant t-Values	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
22	Pseudo Code	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
23	Axis-Parallel Clipping Region	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.

Page	Slide title	How to read it
24	Comparison with Cohen-Sutherland	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
25	5.3 Sutherland-Hodgman Polygon Clipping	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
26	Sutherland-Hodgman Polygon Clipping 1/2	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
27	Sutherland-Hodgman Polygon Clipping 2/2	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
28	Four Cases	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
29	Proceeding	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
30	Pipeline Version	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
31	Pseudo Code	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
32	Inside Test	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
33	Intersection Computation	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
34	5.4 Weiler-Atherton Polygon Clipping	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
35	Weiler-Atherton Polygon Clipping	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
36	Example Run	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
37	Intersection Types	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
38	Contour Processing	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?

Page	Slide title	How to read it
39	Evaluation	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
40	5.5 Greiner-Hormann Polygon Clipping	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
41	Greiner-Hormann	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
42	Chalk Wagon Metaphor	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
43	Contour Detection	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
44	Set Operations	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
45	Data Structure 1/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
46	Data Structure 2/2	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
47	Proceeding	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
48	Pseudo Code	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
49	Winding Number 1/3	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
50	Winding Number 2/3	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
51	Winding Number 3/3	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
52	Special Cases	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
53	Evaluation	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.

Page	Slide title	How to read it
54	by exploiting binary outcodes	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
55	Literature and other sources used in this chapter	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
56	Addison-Wesley 2013.	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?

05.1 Cohen-Sutherland Line Clipping

Source location: Section 5.1

Commentary

Cohen-Sutherland assigns outcodes to line endpoints based on which side of the clipping window they lie on. These codes allow quick accept, quick reject, or partial clipping. The algorithm is efficient because many cases are decided before computing intersections.

The exam-friendly idea is the logic of region codes: if both endpoints are inside, keep the segment. If the bitwise AND of the outcodes is nonzero, both endpoints share an outside region and the segment can be rejected. Otherwise, compute an intersection and continue.

Mental model

Outcodes are a cheap classification before doing intersection math.

Check yourself

What does a nonzero bitwise AND of two endpoint outcodes imply?

05.2 Cyrus-Beck Line Clipping

Source location: Section 5.2

Commentary

Cyrus-Beck treats the line parametrically and clips against convex boundaries. Instead of repeatedly using region codes, it finds entering and leaving parameter values along the line.

The key is to reason about the line parameter t . A clipped line segment is not a new unrelated line; it is a restricted interval of the original parametric line. Boundary tests shrink that interval.

Mental model

Clipping a parametric line means narrowing the valid t interval.

Check yourself

Why does Cyrus-Beck naturally require convex clipping regions?

05.3 Sutherland-Hodgman Polygon Clipping

Source location: Section 5.3

Commentary

Sutherland-Hodgman clips a polygon against one boundary at a time. For each edge of the polygon, it decides whether vertices are inside or outside and emits zero, one, or two vertices depending on transitions across the boundary.

This algorithm is easiest to understand as a stream processor. Feed in a polygon, clip it against the left boundary, feed the result into the right boundary, and so on. Each boundary can add intersection vertices.

Mental model

Polygon clipping is repeated edge-by-edge filtering plus intersection insertion.

Check yourself

When can polygon clipping increase the number of vertices?

05.4 Weiler-Atherton and Greiner-Hormann

Source location: Sections 5.4-5.5

Commentary

These algorithms address more complex polygon clipping scenarios, especially when polygon relationships are not as simple as convex-window clipping. They organize intersections and traversal rules to produce correct output boundaries.

For preparation, focus on why simpler algorithms are not enough. Complex polygon intersection can require following alternating boundaries of subject and clipping polygons. That is a topology problem, not only a line-intersection problem.

Mental model

Complex polygon clipping is about traversing boundary networks after intersections are known.

Check yourself

Why are intersection ordering and traversal rules important for polygon clipping?

End-of-lecture summary

If you remember only one thing from Lecture 05, remember this: This lecture focuses on algorithms that remove or trim geometry against boundaries. Clipping is a geometric operation before rasterization, not a visibility test between overlapping objects.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.